

Hawks, eagles, and relatives

PHYLUM	Chordata
CLASS	Aves
ORDER	Accipitriformes
FAMILIES	4
SPECIES	265

This diurnal group of hunters (sometimes also referred to as raptors) once included the falcons and caracaras, but genetic studies have placed them into a separate group. Although many other birds feed on living animals, these birds of prey are set apart by their remarkably acute eyesight, muscular legs, and

sharp bill and talons—as well as by their aerial ability and sophisticated hunting techniques. This large group includes eagles, hawks, vultures, buzzards, and ospreys, among others. Most birds of prey hunt by day, which distinguishes them from owls. As a group, raptors feed on a large assortment of living animals, from worms and snails to fish, reptiles, amphibians, mammals, and other birds. They also eat carrion. Birds of prey are found almost worldwide but are most common in open country in warm parts of the world.

Anatomy

Birds of prey range in size from the tiny hawk, no larger than a starling, to condors, which have a wingspan of up to 10 ft (3.2 m). Some are broad-winged and heavy, while others are slight and streamlined. Most species have a large head and short neck, although vultures have a long, bare neck that allows them to reach inside carcasses. One of the most distinctive features of the group is the bill: in almost all species, it is powerful and hooked, with sharp edges for tearing flesh. The exact form of the bill varies, reflecting differences in diet (see below, right). A bird of prey's other main tool is its feet, which are strong and muscular with long, sharp claws known as talons (see below, left). The plumage of most species is in subdued colors (such as brown, gray, black, or blue), often combined with white.



TALONS
Most birds of prey kill with their talons rather than their bill. The talons pierce vital organs and break slender bones. Three of the 4 toes point forward, and one points backward. In many species (including this white-tailed sea eagle), the backward-pointing claw is the longest and sharpest. It gives the foot a powerful pincerlike action for holding and crushing prey that can be difficult for the bird to release.



BILLS
The distinct hook and sharp edges of a bald eagle's bill are typical of many birds of prey. Variations on this shape can be seen in other species. Hawks have a toothlike ridge in the upper bill for breaking the spine of their prey, while the snail kite uses its long, hooked bill to pull snails from their shells.

Senses

Birds of prey hunt mainly by sight, and they are renowned for their acute vision. It is estimated that they can see at least 4 times as much detail as a human, an ability that helps them to pick out prey from a great distance. This arises from having a high concentration of cone cells in the retina, and also from having relatively large eyes. Some species have excellent hearing, notably the harriers, which hunt over thick vegetation. A few vultures also have an exceptional sense of smell, by which they can locate hidden carrion.



Hunting

Catching live prey takes much practice and often has a high failure rate. Adult birds are usually better hunters than young ones, but even adults miss more often than they strike. Some species vary their technique according to their prey—buzzards, for example, hover over small mammals but search for earthworms on the ground. Others, such as the osprey, are more specialized (see below). Most birds of prey hunt alone, but in a few species (including some eagles) pairs work together, one bird flushing out the prey and the other striking.



FISHING
The osprey feeds almost entirely on fish and has a unique catching technique. It approaches in a low-angled dive and throws its long legs forward to grab a fish with its talons. To help it grip, the osprey may reverse one of its 3 forward-pointing toes.



EYE PROTECTION
Like many birds of prey, the goshawk has a ridge over each eye, which provides shade and protects the eye from struggling prey. In many species, the eye is also protected by a transparent third eyelid (nictitating membrane).